



UNIVERSITI TUNKU ABDUL RAHMAN

Lee Kong Chian Faculty of Engineering and Science

Bachelor of Science (Honours) Software Engineering

UECS2333 Human Computer Interaction Design

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Assignment Task 1

No.	Name	Student ID	Practical Group
1	Ang Yong Xen	2301521	P9
2	Kyle Yamato Christian	2302452	P9
3	Tan Jun Yin	2302533	P10
4	Yap Kai Zheng	2302943	P3

Total Marks:

1	Techniques to get Requirements		10%
2	Persona Development		10%
3	Scenario Development (Functions)		5%
4	Scenario Development (Descriptions)		10%
Total			35%

Description of techniques

To understand our users, their motivations and their tasks, we conducted a comprehensive study using questionnaire and interview. Our primary goal was to gather detailed information that would help to improve our system development and enhance user satisfaction. With the data collected, we have gained a deeper understanding of user requirements when utilizing the reservation system on the mobile platform.

The questionnaire was designed to cover various aspects of user interaction, including demographic information, their difficulties in using the booking system and their expectation for the new booking system. It consisted of 20 questions, including multiple-choice, open-ended question and free-response question, these questions are to capture quantitative and qualitative data from the users. Also, the questionnaire was distributed online through Google Forms. We received responses from 30 users, representing a diverse cross-section of our user base in terms of age, gender, and diverse background.

In addition, qualitative responses identified specific pain points, including the absence of customer service and frequent system bugs, which contribute to the difficulty of using the system.

After gathering the questionnaire results, we conducted semi-structured interviews with 5 users from our 30 respondents to gather in-depth qualitative data on their experiences, needs, and preferences related to booking bowling lanes.

The results have provided valuable insights into our user base and system. These data indicate that while users appreciate the convenience and cost-effectiveness of our booking system, there are opportunities for improvement in navigation and customization. These findings will help guide our next new system development, allowing us to address user needs more effectively.

Results of Techniques

Questionnaires:

1) Demographic Information:

- Majority of respondents are between the ages of 22-25 (40%) and 26-30 (36.7%).
- Gender distribution is fairly even with (53.3%) female and (46.7%) male.

2) Bowling Frequency:

- 43.3% of respondents go bowling occasionally, while 30% go monthly.

3) Preferred Booking Time:

- Evening is the most preferred time for booking (36.7%), followed by late night (26.7%).

4) Bowling Companions:

- Respondents usually bowl with family (46.7%) and friends (46.7%).

5) Booking Devices:

- Smartphones are the most used device for booking (53.3%), followed by tablets (26.7%) and laptops (23.3%).

6) Technical Issues:

- The most common issues include a complicated user interface (40%) and slow performance/loading times (30%).

7) Online Payments:

- (70%) of respondents find online payments efficient and convenient.

8) Rescheduling and Cancellation:

- (63.3%) of respondents find it difficult to reschedule or cancel a booking.

9) Information Sufficiency:

- (43.3%) of respondents agree that the current system provides sufficient information about available time slots and lanes.

10) Ease of Use:

- (46.7%) find the current system understandable and easy to use.

11) Customer Service Satisfaction:

- (43.3%) are satisfied with the customer service provided by the current system.

12) Payment Process Satisfaction:

- (53.3%) are satisfied with the payment process.

13) Overall Experience:

- (43.3%) rate their overall experience with the current system as good.

14) Difficulties Faced:

- Common difficulties include the system not being user-friendly, missed reservations, no reminders, and no customer support.

15) Desired Features for New System:

- Respondents expect features like easy navigation, a quick booking process, secure payment options, detailed booking confirmation, the ability to cancel/change bookings, and live customer service.

16) Booking Speed:

- Majority expect the booking process to take less than 1 minute (36.7%) or less than 5 minutes (26.7%).

17) Booking Confirmation:

- Most respondents prefer to receive booking confirmation via in-app notifications (46.7%) and email (26.7%).

18) Additional Features:

- Desired additional features include booking for events/parties (30%), booking multiple lanes (36.7%), integration with calendar apps (36.7%), and loyalty programs (26.7%).

19) User-Friendly Interface:

- (26.7%) strongly agree that the new system should have a more user-friendly interface.

20) Suggestions or Expectations towards the New Bowling Booking System:

- Offer food options as an add-on when making a reservation.
- Implement a bug-free system.
- Provide real-time customer support.

Interview:

Question 1: Typical Experience When Booking a Bowling Lane

- **Interviewee 1:** Uses the website or app, finds it outdated with occasional booking conflicts due to real-time update issues. Wishes for more advanced filtering options.
- **Interviewee 2:** Uses the mobile app and finds it convenient but experiences crashes during peak hours. Desires a more intuitive interface and better control over lane selection.
- **Interviewee 3:** Uses the mobile app, finds it convenient but not very intuitive with slow loading times and too many steps.
- **Interviewee 4:** Prefers calling the bowling alley directly to book, finds it time-consuming if the line is busy. Website use is straightforward but lacks features like easy cancellation.
- **Interviewee 5:** Books online, finds the website confusing and not user-friendly with difficulties in finding available slots and uncertainty if bookings went through.

Question 2: Useful or Frustrating Features

- **Interviewee 1:** Finds the real-time calendar view useful but frustrating due to outdated availability and limited customization options.
- **Interviewee 2:** Appreciates saving preferred lanes and times but is frustrated with app crashes and unreliable performance.
- **Interviewee 3:** Values quick payment options but finds the login process frustrating and the booking steps excessive.
- **Interviewee 4:** Finds booking confirmation emails useful but is frustrated by the lack of online rescheduling or cancellation options.
- **Interviewee 5:** Likes discount notifications but is frustrated by the slow, crash-prone system and stressful booking process.

Question 3: Desired Additional Features or Improvements

- **Interviewee 1:** Desires real-time updates, a streamlined booking process, advanced filtering options, better integration with calendar apps, and an in-app chat feature.
- **Interviewee 2:** Wants integration with scheduling tools, real-time updates, a more intuitive interface, multiple lane booking for events, and advanced lane filtering.
- **Interviewee 3:** Seeks calendar app integration, a simplified booking interface, better performance, and faster load times.
- **Interviewee 4:** Would appreciate multiple lane booking, a loyalty program, and easy online cancellation and rescheduling options.
- **Interviewee 5:** Desires a more user-friendly interface, features to compare prices and availability at different alleys, a reliable payment system, and straightforward booking confirmation.

Persona 1 – Novice User



Jasmine Lim

Age: 24 years

Location: Urban area with access to several bowling alleys

Sex: Female

Education Level: Bachelor's Degree in Graphic Design

Marital status: Single

Occupation: Junior Graphic Designer

BIOGRAPHY

Jasmine is a 24-year-old junior graphic designer who enjoys exploring new hobbies in her free time. Recently, her friends suggested trying out bowling as a fun group activity. Jasmine is excited but a little apprehensive about the booking process since she's never booked a lane before. She values a straightforward, user-friendly system that allows her to easily see lane availability, make bookings, and understand pricing.

BEHAVIORAL INFORMATION

- New to bowling, books occasionally with friends
- Often books for small groups of friends
- Prefers weekends or evenings after work
- Likely to become a repeat customer if the experience is positive and user-friendly

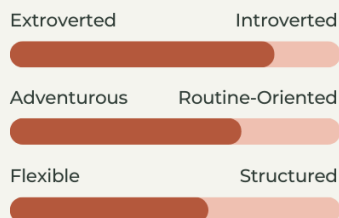
GOALS

- To have an enjoyable and hassle-free experience booking bowling lanes
- Excited to try out bowling as a new recreational activity with friends, values convenience and ease of use

MOTIVATION

Socializing with Friends	★★★★★★
Exploring New Activities	★★★★★★
Convenience and Ease of Use	★★★★★★
Work-Life Balance	★★★★★★

PERSONALITY



PREFERENCE

- Primarily uses a smartphone for most activities, occasionally uses a laptop for more detailed tasks.
- Prefers mobile apps for their convenience and accessibility, but is also comfortable using websites.
- Familiar with common apps and online booking systems but values simplicity and ease of use.
- Likes apps and systems that have intuitive interfaces, quick loading times, and minimal steps to complete tasks.
- Cares about data privacy and security, especially when making online payments.

"I just want to see available lanes and book them in a few clicks. It shouldn't be this hard to organize a fun night out with friends."

Scenario 1 – Novice User

Jasmine Lim is a 24-year-old junior graphic designer who enjoys exploring new hobbies in her free time. She plans to try out bowling as a new hobby with her friends during her weekends. She is excited to bowl with her friends by booking a reservation during the evenings after work on weekends. She needs an application to check the availability of the bowling lanes, make bookings, and understand the pricing. Therefore, she decided on a straightforward, user-friendly system that allows her to book a bowling reservation for the first time.

Jasmine found a new software application called BowlZone, available on the App Store that she had not used before. Based on the introduction and brief overview of the application's features, it satisfied the requirements outlined by Jasmine. Consequently, she decided to download it and create an account. After Jasmine downloaded the application, she opened it and read the privacy statement, terms, and conditions carefully to continue to use it. Then, she signed up for a new account using her phone number for verification for the first time. She also created a new profile picture and username.

Jasmine is a first-time user of this application. She wants to know more about the application's functions, so she decides to click on the 'Learn More' button to have further information and explanation. Furthermore, she presses the 'My Rewards' button at the 'Home' page to check where there is a promotion or any membership discount. She reads all the messages and the procedures for how to sign up for a membership account. After that, she decides to book a reservation for bowling before considering whether she wants to join the membership.

Jasmine makes a first booking by clicking the 'Book Now' button on the bottom first of the 'Home' page. She can view all the available times and bowling lanes. After choosing the time and the bowling lanes, Jasmine presses the 'Confirm Booking' button to proceed with the payment methods. Before proceeding to the payment, the system will require her to double-check the booking details again to ensure all the information is correct and accurate. Next, she can choose the payment methods that are available in the application, such as credit or debit card, Touch-N-Go Ewallet, online banking, or cash payment at the bowling alley counter. If she chooses the cash payment method, there is a condition where she needs to pay a 30% deposit and the remaining balance can be paid at the bowling alley counter.

After Jasmine makes a bowling booking, the booking details will be stored on the 'My Booking' page, which is the second button from the left at the bottom. She can access the booking details by pressing the 'My Booking' button. Also, she will receive booking points for redeem discount voucher for the next booking, she can redeem it at 'My Rewards' section. To exit the page, she can press 'x' on the top right-hand side, which will return her to the 'Home' page. If she encounters any problems or issues regarding the booking details, she can press the second button from the right at the bottom, which is 'Contact Us', to get further help and support from the bowling alley staff. Additionally, before the bowling time begins, the system will notify Jasmine through the application with a message that says, "Your bowling reservation begins in 30 minutes. Kindly get ready!"

Mike Paradis

Bio

Mike is a dedicated engineering professional with a passion for bowling. He is an expert in the bowling game, playing weekly and keen on improving his skills. He also has won a championship in a local competition and received a prize of RM1500. He often takes friends to the bowling center and is eager to make new friends and exchange techniques. At 26, he focuses on optimizing his leisure time by booking bowling matches efficiently. He strives to find a bowling center that meets the needs of his entire group of friends, ensuring a comfortable and enjoyable experience for all. His priorities in the booking process are price, comfort, and speed, seeking a seamless and convenient experience from start to finish.



Age: 26

Sex: Male

Status: Single

Location: Kuala Lumpur, Malaysia

Occupation: Engineering Professional

Goals

- To book a bowling match quickly.
- To find a bowling centre that satisfies the needs of the entire group of friends.
- To feel quick and convenient from the beginning of the booking experience to the end.

Frustration

- Too many irrelevant ads
- Poor interface design
- Multiple steps to book reservation

Favourite Bowling Centre



Personality

Extrovert

Motivation

Friendly

Passionate

Effort

Motivation

Price



Comfort



Eco-friendly



Social



Personalization



"I want a bowling center that can meet the needs of me and my friends, ensuring that everyone can enjoy a comfortable and pleasant experience."

Scenario 2 – Expert User

Mike Paradis is a 26-year-old professional engineer who has been an avid bowler for over ten years. Recently, Mike won a championship in a local bowling tournament, earning a prize of RM1500. Motivated by this achievement, he wants to keep improving his performance and maintain his best form through consistent training. Mike regularly books bowling lanes to practice with his friends too and prefers using advanced features of bowling reservation systems to ensure he maximizes his time at the centre.

Mike discovered BowlZone a few months ago and has been using it extensively since then. He appreciates the application's efficiency and the range of options it offers. Mike logs in to BowlZone using his face ID for quick and secure access. His profile is already set up with all his preferred settings, including his favourite bowling lanes and usual booking times.

Mike wants to book a lane for his next practice session with his friends. He starts by opening the application and quickly navigates to the 'My Booking' page, where he can see all his upcoming and past reservations. He clicks on the 'Quick Book' option, which automatically fills in his preferred times, lanes and number of pax based on his previous bookings. Mike reviews the details and confirms the reservation with a single tap, using his saved credit card for the payment. After payment is confirmed, Mike will receive extra member points as an expert user and these points will save into 'My Rewards' section. Considering his expertise with the application, Mike often utilizes the 'Advanced Settings' feature to customize his bookings. This feature includes selecting specific lane conditions like lane surface and adding any additional equipment for Mike and his friend might need such as bowling shoes or balls.

Mike also keeps track of his performance through the 'Statistics' feature, which provides detailed analytics on his games. After each session, he updates his scores and reviews his progress over time. If Mike encounters any issues or needs assistance, he prefers using the in-app chat feature found under 'Contact Us', which connects him directly with a support representative for quick resolution for his issues. He also stays informed about promotions and discounts through push notifications and regularly checks the 'My Rewards' section to see if he has earned any new rewards. He can also redeem discount vouchers for his next booking using the points accrued from a previous reservation. After Mike has completed his booking, he exits the booking application. Additionally, before his practice session begins, Mike receives a notification reminding him of his booking and providing any relevant updates, such as changes in lane availability or special offers.

Appendixes

Questionnaire – Questions Section

Part A – Demographic data

1. What is your age? *

☐ 18-21

☐ 22-25

☐ 26-30

2. What is your gender? *

☐ Male

☐ Female

3. How often do you go bowling? *

☐ Weekly

☐ Monthly

☐ Occasionally

☐ Rarely

☐ Never

4. What time of day do you prefer to book a bowling reservation? *

☐ Morning

☐ Afternoon

☐ Evening

☐ Late night

5. Who do you usually bowl with? *

☐ Family

☐ Friends

☐ Alone

☐ Other: _____

6. What devices do you use when booking bowling reservations? *

☐ Smartphone

☐ Tablet / iPad

☐ Laptop

☐ Other: _____

Part B – Difficulties when using the booking system

7. Do you encounter any technical issues or difficulties when using the **current** reservation system? *

- ☐ Login / sign in problems
- ☐ Complicated user interface
- ☐ Limited availability time slots
- ☐ Slow performance / slow loading time
- ☐ No / delayed confirmation
- ☐ Payments security
- ☐ Poor customer services
- ☐ Other: _____

8. Do you think online payments are efficient and convenient to use? *

- ☐ Yes
- ☐ No

9. Do you find it easy to reschedule or cancel a bowling booking in the **current** system? *

- ☐ Yes
- ☐ No

10. The **current** bowling booking system provides sufficient information about the available time slots and lanes. *

						
STRONGLY DISAGREE	DISAGREE	NEUTRAL	AGREE	STRONGLY AGREE		
	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

11. The **current** bowling booking system is understandable and easy to use compared to others. *

						
STRONGLY DISAGREE	DISAGREE	NEUTRAL	AGREE	STRONGLY AGREE		
	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

12. How satisfied are you with the customer services of you **current** bowling booking system? *

						
★ ★ ★ ★ ★	★ ★ ★ ★ ★	★ ★ ★ ★ ★	★ ★ ★ ★ ★	★ ★ ★ ★ ★		
	1	2	3	4	5	
Highly Dissatisfied	☐	☐	☐	☐	☐	Highly Satisfied

13. How satisfied are you with the payment process through the **current** reservation system? *

						
★ ★ ★ ★ ★	★ ★ ★ ★ ★	★ ★ ★ ★ ★	★ ★ ★ ★ ★	★ ★ ★ ★ ★		
	1	2	3	4	5	
Highly Dissatisfied	☐	☐	☐	☐	☐	Highly Satisfied

14. How would you rate your overall experience with the **current** bowling booking system? *

worst	poor	average	good	excellent
				
★ ★ ★ ★ ★	★ ★ ★ ★ ★	★ ★ ★ ★ ★	★ ★ ★ ★ ★	★ ★ ★ ★ ★
1	2	3	4	5
☐	☐	☐	☐	☐

15. Do you have any difficulties when using the current bowling booking system?
Justify your answer.

Your answer

Part C – Expectation for the new booking system

16. What features do you expect in **new** bowling booking system? *

- ☐ Easy navigation
- ☐ Quick booking process
- ☐ Secure payment options
- ☐ Detailed booking confirmation
- ☐ Availability to cancel and change bookings details
- ☐ Live customer service and support
- ☐ Other: _____

17. What is your expectation regarding the speed of the booking process? *



- ☐ Less than 1 minute
- ☐ Less than 5 minutes
- ☐ Less than 10 minutes
- ☐ More than 10 minutes

18. How do you expect to receive the booking confirmation? *



- ☐ Email
- ☐ SMS
- ☐ Whatsapp
- ☐ In-app notification
- ☐ Others

19. What other additional features would you like to see in the **new** bowling booking system? *



- ☐ Booking for events and parties
- ☐ Booking multiple lanes
- ☐ Integrating with calendar apps
- ☐ Loyalty programs and rewards
- ☐ Other: _____

20. Do you agree the **new** bowling should have a more user-friendly interface? *



1 2 3 4 5

Strongly Disagree ☐ ☐ ☐ ☐ ☐ Strongly Agree

21. Do you have any specific suggestion or expectations for the new bowling booking system that are not mentioned above?

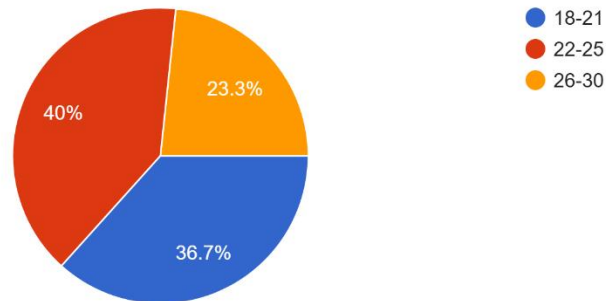
Your answer

Questionnaire – Results Data

Part A – Demographic Data

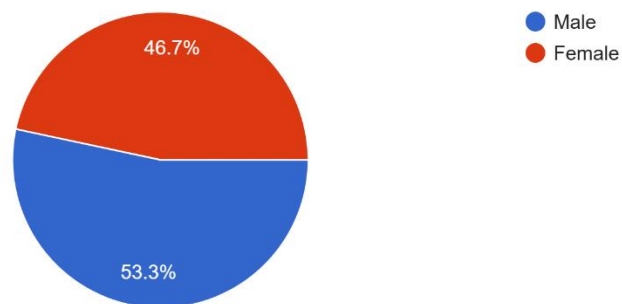
1. What is your age?

30 responses



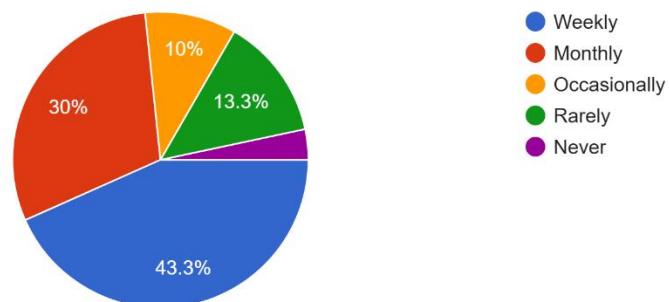
2. What is your gender?

30 responses



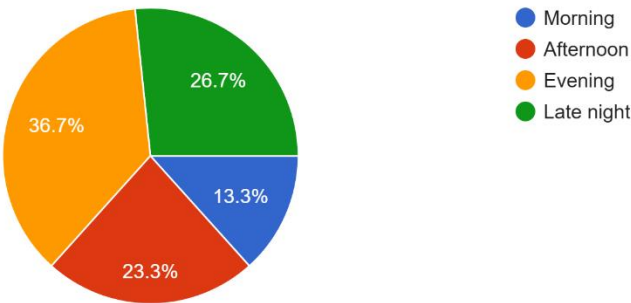
3. How often do you go bowling?

30 responses



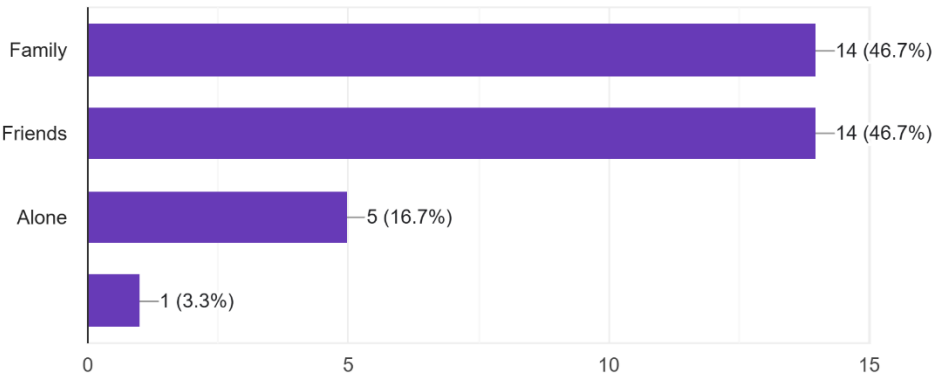
4. What time of day do you prefer to book a bowling reservation?

30 responses



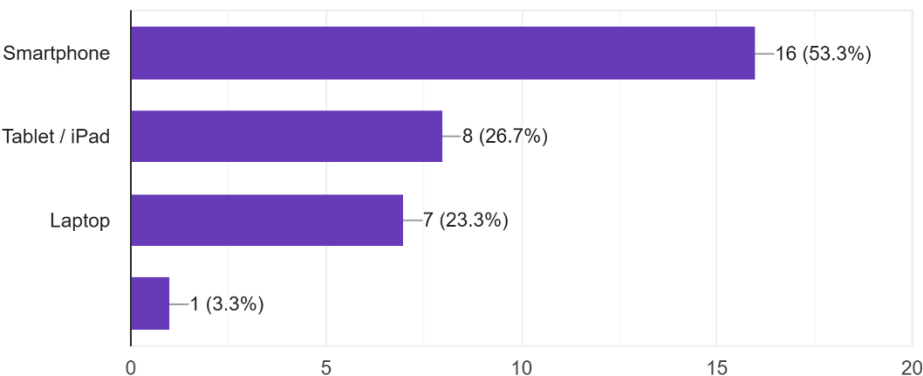
5. Who do you usually bowl with?

30 responses



6. What devices do you use when booking bowling reservations?

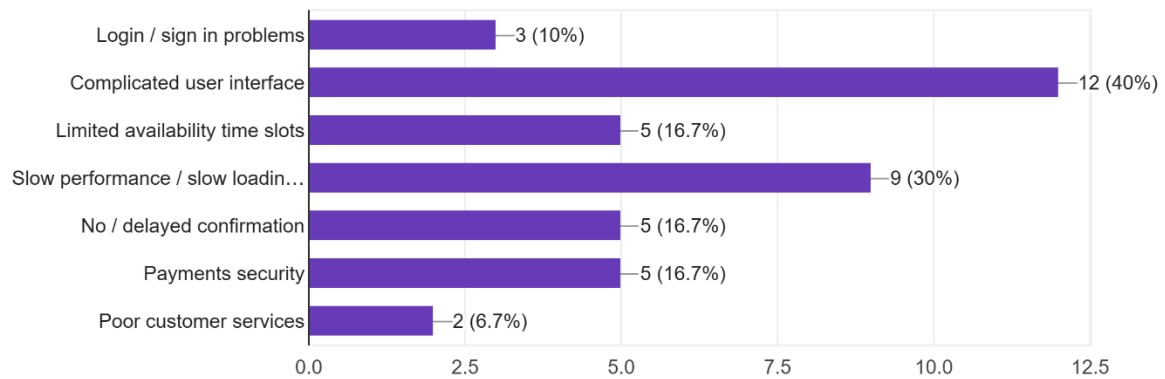
30 responses



Part B – Difficulties in using booking system

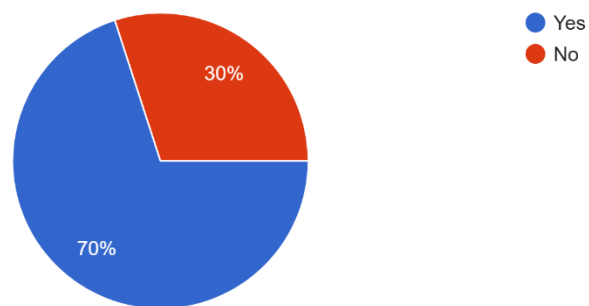
7. Do you encounter any technical issues or difficulties when using the current reservation system?

30 responses



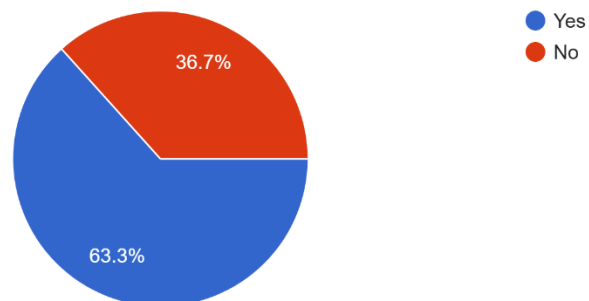
8. Do you think online payments are efficient and convenient to use?

30 responses



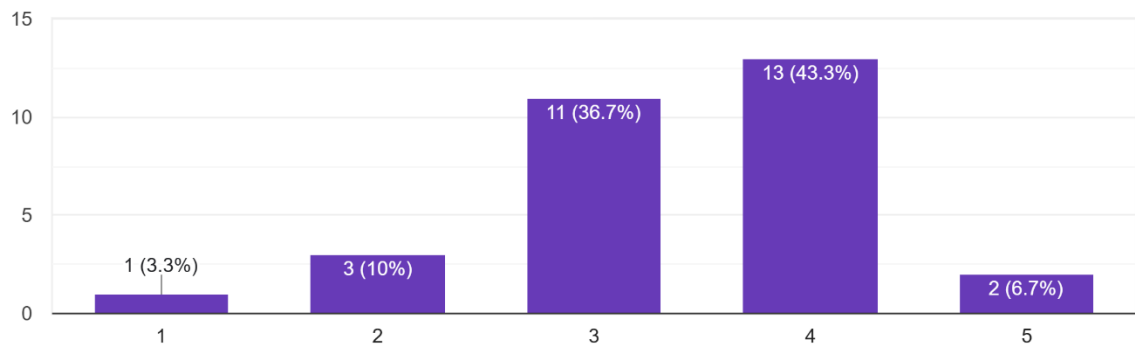
9. Do you find it easy to reschedule or cancel a bowling booking in the current system?

30 responses



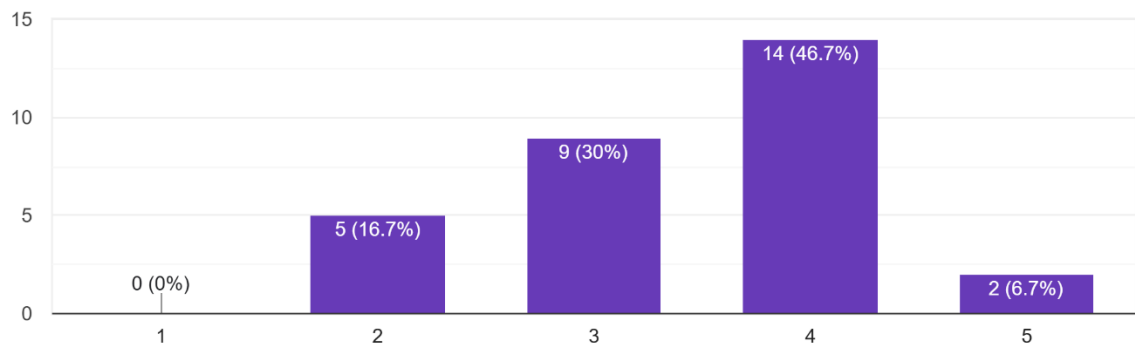
10. The current bowling booking system provides sufficient information about the available time slots and lanes.

30 responses



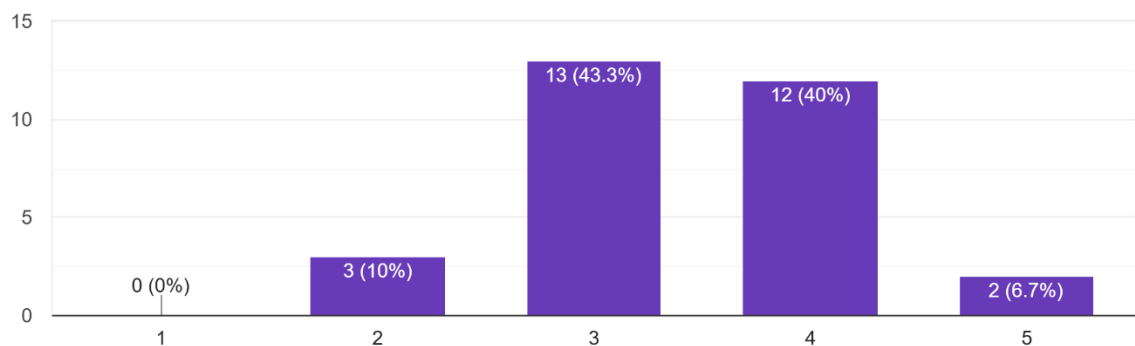
11. The current bowling booking system is understandable and easy to use compared to others.

30 responses



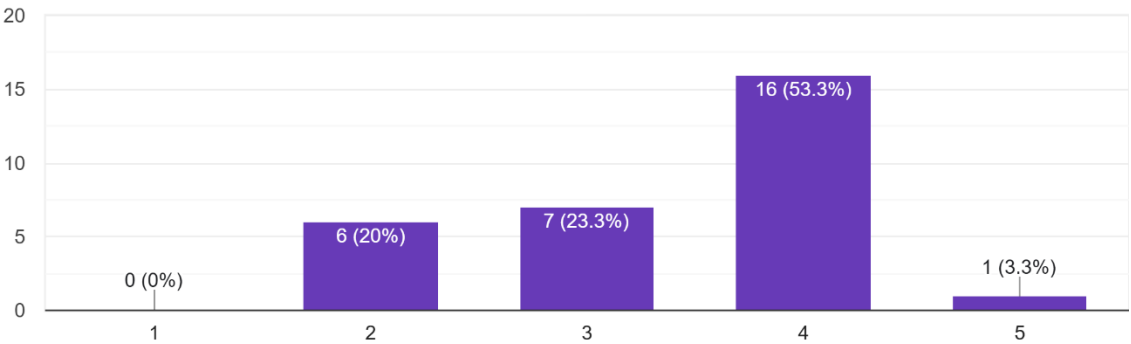
12. How satisfied are you with the customer services of you current bowling booking system?

30 responses



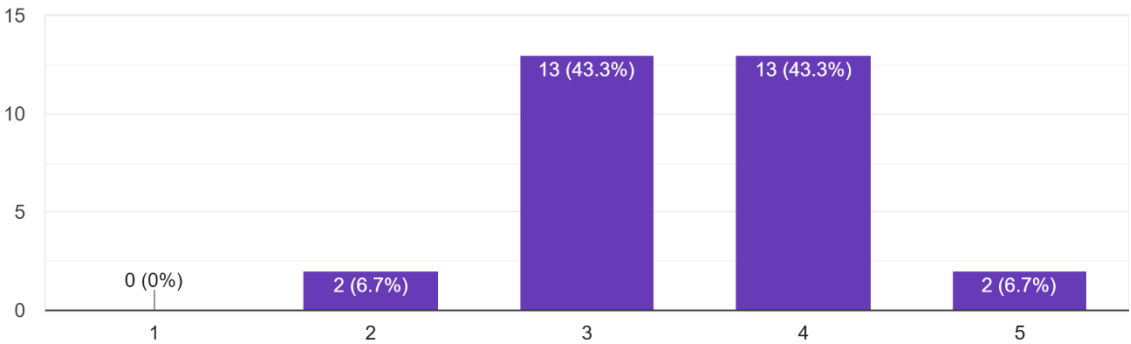
13. How satisfied are you with the payment process through the current reservation system?

30 responses



14. How would you rate your overall experience with the current bowling booking system?

30 responses



15. Do you have any difficulties when using the current bowling booking system? Justify your answer.

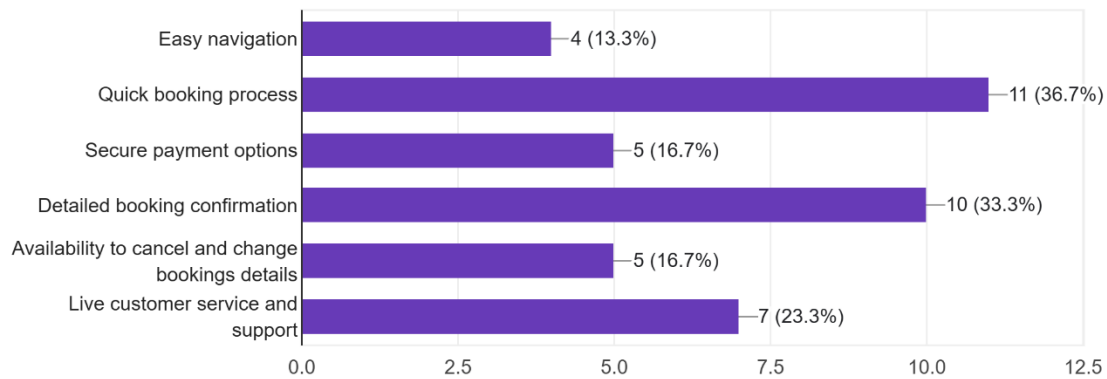
12 responses

No
Missed Reservations
It is not user-friendly
No difficulty
It does no send me a remainder of my booking.
The system is buggy
No customer support
none in particular
complicated interface

Part C – Expectation for the new booking system

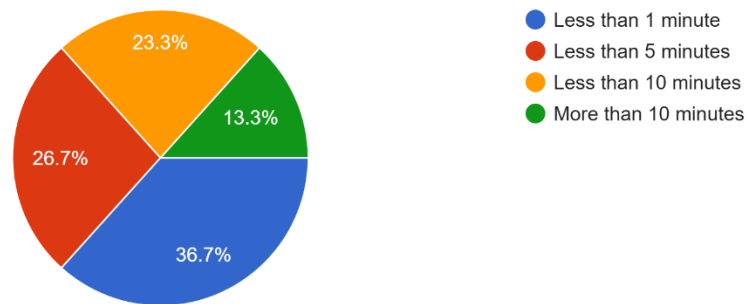
16. What features do you expect in new bowling booking system?

30 responses



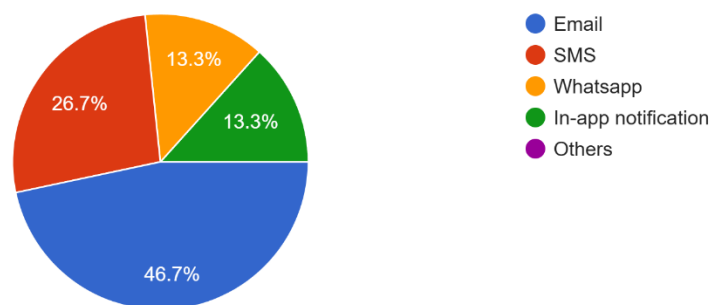
17. What is your expectation regarding the speed of the booking process?

30 responses



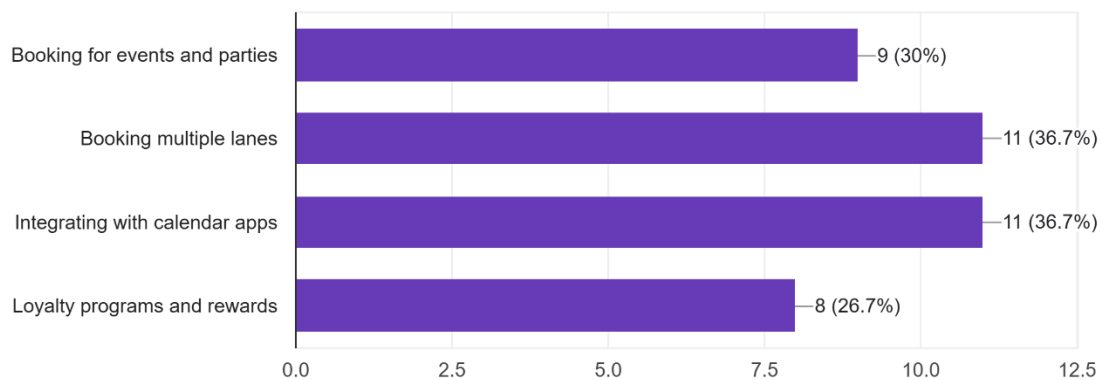
18. How do you expect to receive the booking confirmation?

30 responses



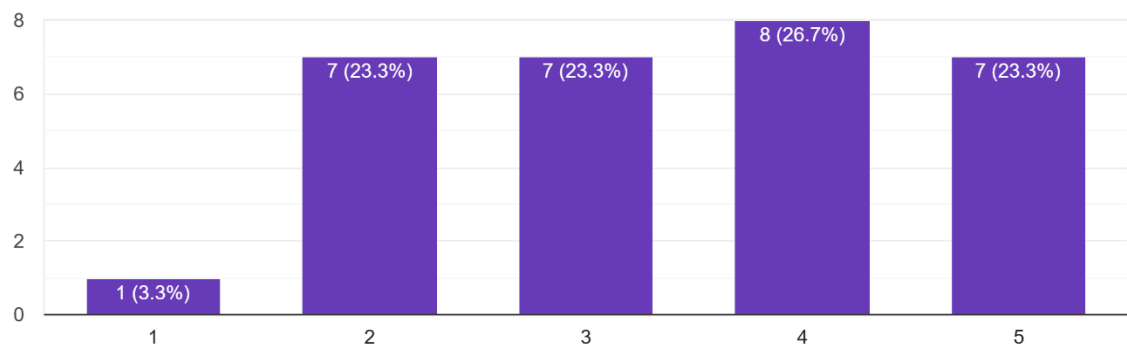
19. What other additional features would you like to see in the new bowling booking system?

30 responses



20. Do you agree the new bowling should have a more user-friendly interface?

30 responses



21. Do you have any specific suggestion or expectations for the new bowling booking system that are not mentioned above?

4 responses

Offer food for as add-on when reserve the bowling spot

No

Make sure the system is not buggy

have a real-time customer support system.

Interview Section

Interview Question:

Question 1: Can you describe your typical experience when booking a bowling lane?

Question 2: What features do you find most useful or frustrating in your current booking system?

Question 3: What additional features or improvements would you like to see in a new bowling booking system?

Interviews Detailed Response:

Question 1: Can you describe your typical experience when booking a bowling lane?

Interviewee 1:

- "I use the bowling alley's website or app, which I find to be somewhat outdated. I can usually find available slots quickly, but the system often doesn't update in real time, causing booking conflicts. I also wish there were more advanced options for filtering and sorting available lanes."

Interviewee 2:

- "I usually book through the bowling alley's mobile app. It works fine most of the time, but I've encountered issues with the app crashing during peak hours. Additionally, the user interface could be more intuitive, and I'd like more control over selecting specific lanes."

Interviewee 3:

- "I primarily use a mobile app for booking. It's convenient but often has a lot of steps to go through. The interface is not very intuitive, and I sometimes experience slow loading times. However, the basic functionality works for me."

Interviewee 4:

- "I usually book by calling the bowling alley directly. This allows me to confirm details instantly, but it can be time-consuming if the line is busy. When I do use the website, it's generally straightforward, though it sometimes lacks key features like easy cancellation."

Interviewee 5:

- "I mostly book online using the bowling alley's website. I find it confusing because the site is not very user-friendly, and I often have trouble finding available slots. The process feels overwhelming, and I'm not always sure if my booking went through."

Question 2: What features do you find most useful or frustrating in your current booking system?**Interviewee 1:**

- "The real-time calendar view of available slots is very useful, but I find it frustrating that the availability isn't always up-to-date. I also wish there were more detailed options for customizing my booking, like lane preferences and booking for groups."

Interviewee 2:

- "I appreciate the option to save my preferred lanes and times, but it's frustrating when the app crashes or the interface doesn't load properly. Real-time updates and more reliable performance are needed."

Interviewee 3:

- "I appreciate the quick payment option, but the login process can be frustrating as it frequently forgets my credentials. I also find the number of steps required to complete a booking excessive and sometimes confusing."

Interviewee 4:

- "I like the booking confirmation emails as they provide all necessary details. However, I get frustrated with the lack of an online option for rescheduling or cancelling bookings. I often have to call, which is inconvenient."

Interviewee 5:

- "I find the discount notifications useful, but the overall system is slow, and I frequently encounter crashes. The website doesn't always load properly, which makes the booking process stressful."

Question 3: What additional features or improvements would you like to see in a new bowling booking system?

Interviewee 1:

- "I'd like to see real-time availability updates and a more streamlined booking process. Advanced filtering options and better integration with other services, such as calendar apps and reminders, would be great. An in-app chat feature for quick support would also be valuable."

Interviewee 2:

- "Integration with other scheduling tools, real-time updates, and a more intuitive interface would be major improvements. I'd also like to see an option to book multiple lanes for events and advanced filtering for lane preferences."

Interviewee 3:

- "Integration with calendar apps to automatically add my bookings would be helpful. Simplifying the booking interface to reduce the number of steps required would improve the experience. Better performance and faster load times are also important."

Interviewee 4:

- "Features like booking multiple lanes at once and having a loyalty program would be appreciated. Additionally, providing easy online options for cancellations and rescheduling would make managing bookings much more convenient."

Interviewee 5:

- "I'd like a more user-friendly interface with clearer instructions. Features to compare prices and availability at different bowling alleys would be helpful. Also, a reliable payment system and straightforward booking confirmation would reduce my stress."

Interview Session Photos:



Interviewee 4



Interviewer



Interviewee 5



Interviewer

